

SWASTH **Maternal Health**

AI-Powered Multilingual Long-Term Care Companion, Hyperlocal Birth Clubs & Frontline Health Continuity Platform

CHALLENGE TRACK

Maternal & Women's Health

PRIMARY USER

Patient & Family Caregivers

CORE USE CASE

Long-Term Care Engagement

The Unseen Crisis: Why Long-Term Maternal Care Breaks Down



Post-Discharge Drop-Off Cliff

Managing maternal health requires **1,000+ continuous days** across pregnancy, institutional delivery, and 7 infant vaccine cohorts.

- **30% Drop-Out:** Mothers disengage after hospital discharge due to zero daily touchpoints.
- **Late Mortality Risk:** Over 50% of maternal complications occur at home during the 42-day postpartum period.
- **Vaccine Gaps:** Severe immunization drop between the 14-week and 9-month doses.

PRIMARY VULNERABILITY: CONTINUITY CLIFF



Administrative Friction & Isolation

Caregiver dropouts stem from logistical overload, fragile paper records, and lack of nearby peer empathy circles.

- **Fragile Paper MCP Cards:** Easily misplaced, stained, or damaged across seasonal migrations.
- **Logistical Overload:** 18+ separate clinical visits, lab tests, ultrasound scans, and welfare schemes.
- **Family Disconnect:** Spouses and elders are completely excluded from active care synchronization.

PRIMARY FRICTION: PAPERWORK & LOGISTICS



The "Cold Clinical App" Failure

Standard healthcare apps suffer **85%+ abandonment within 30 days** because they are cold and transactional.

- **Zero Daily Pull:** Mothers never open an app solely to view a monthly appointment reminder.
- **English & Jargon:** Complex clinical terms alienate vernacular and rural families.
- **Culturally Disconnected:** Missing daily bonding, Garbh Sanskar rituals, and peer community warmth.

PRIMARY HURDLE: HABIT DISCONNECTION

The "Hook & Bridge" Model: Sustaining Daily Engagement

THE DAILY ENGAGEMENT HOOK

Garbh Sanskar Rituals & Hyperlocal Birth Clubs

 **4Q Holistic Daily Activities:** 12 daily micro-habits across IQ (cognitive stimulation), EQ (emotional bonding), PQ (prenatal mobility), and SQ (spiritual calming).

 **Hyperlocal Birth Clubs:** Cohort circles connecting mothers in the same delivery month & PHC radius for mutual peer support and comfort routines.

 **Garbh Samvaad & Raga Audio:** Soothing audio affirmations, baby bonding conversations, and calming prenatal sounds in regional languages.

 **Community Moderation:** NVIDIA NeMo Guardrails intercept folk medical claims, keeping peer circles empathetic, positive, and safe.

 **Daily Habit Hook:**

82% Daily Active User Retention Target Across 1,000 Days

THE HEALTHCARE CONTINUITY BRIDGE

Vaccine Milestones, Diet Charts & Caregiver Sync

 **Smart Vaccine Cohort Tracker:** Automates 6, 10, 14-week and 9-month immunization scheduling with vernacular audio prompts.

 **Doctor-Approved Weaning Diet:** Stage-wise infant nutrition guides and maternal recovery lactation plans approved by pediatricians.

 **Caregiver Multi-User Sync:** Partners and grandmothers receive audio/WhatsApp reminders in their native tongue with explicit consent.

 **24/7 Offline Sync:** Complete operation without internet; automated background push to PostgreSQL once connectivity resumes.

 **Clinical Bridge:**

100% Adherence to National Immunization Schedule

Full Alignment with SWASTH's 6 Core Pillars

01 · CARE JOURNEY COMPANION

Stage-Aware Guidance

Adaptive guidance tailored dynamically from conception to 24-month toddlerhood.

- ✓ Trimester checklists & hospital delivery bag planner
- ✓ Week-by-week baby development & fruit-size analogies
- ✓ 42-day postpartum recovery & lochia tracking protocols
- ✓ 24/7 offline local caching with zero internet dependency

Operational Impact: Automated vernacular alerts; ANMs access synchronized due-lists during VHSND sessions.

Target Metric

95% Milestone Awareness

02 · LONG-TERM CARE ENGAGEMENT

Sustained Multi-Year Adherence

Overcomes app churn through cultural bonding, peer circles, and habit streaks.

- ✓ Daily 4Q Garbh Sanskar micro-routines (IQ, EQ, PQ, SQ)
- ✓ Gamified streak badges & joyful milestone celebrations
- ✓ Hyperlocal Birth Clubs for empathetic neighborhood bonding
- ✓ Vernacular audio lullabies & partner bonding discussions

Operational Impact: Replaces episodic alerts with daily micro-habits, maintaining caregiver engagement over 1,000 days.

Target Metric

>80% 90-Day Retention

03 · FAMILY & CAREGIVER SUPPORT

Caregiver Synchronization

Dual-consent sync reducing the mental load on expectant and new mothers.

- ✓ Partner Sync: Husbands receive tailored preparation checklists
- ✓ Vernacular voice nudges for grandmothers and family elders
- ✓ Shared emergency transport & blood donor contact planning
- ✓ WhatsApp notification mirrors for non-smartphone users

Operational Impact: Distributes logistical load across the household while respecting patient data consent boundaries.

Target Metric

3x Caregiver Participation

04 · HEALTHCARE NAVIGATION & ACCESS

Clinic & Facility Guidance

Demystifies the clinical path across token counters, lab tests, and referrals.

- ✓ Automated ultrasound window alerts & fasting instructions
- ✓ PHC-to-District hospital seamless referral handoff notes
- ✓ Pre-filled check-in questions to optimize doctor consultation
- ✓ Token counter & registration window procedural walkthroughs

Operational Impact: Eliminates wasted clinic trips and guides patients step-by-step through government facility registration.

Target Metric

40% Shorter Clinic Wait Time

05 · FINANCIAL & ADMIN SUPPORT

Maternity Scheme Navigator

Simplifies government welfare schemes so financial delays never block care.

- ✓ Step-by-step documentation tracker for PMMVY (₹5,000 grant)
- ✓ JSY & JSSK cash transfer entitlement calculators
- ✓ Instant digital linkage to ABHA ID & Ayushman Bharat
- ✓ Offline photo scanner for bank passbook & Aadhaar linkage

Operational Impact: Digitizes eligibility verification and bank account seeding so expectant mothers receive cash transfers on time.

Target Metric

100% Entitlement Realization

06 · PATIENT HEALTH JOURNEY PROGRESS

Longitudinal Milestone View

Single-pane interactive timeline of mother and child developmental health.

- ✓ Complete digital MCP card sync with permanent cloud backup
- ✓ Visual status of 7 infant vaccine cohorts & growth percentiles
- ✓ Verified doctor visit history & diagnostic summary archive
- ✓ WHO-compliant child weight, height, and head growth curves

Operational Impact: Unifies fragmented prenatal, intrapartum, and child immunizations into an immutable, longitudinal timeline.

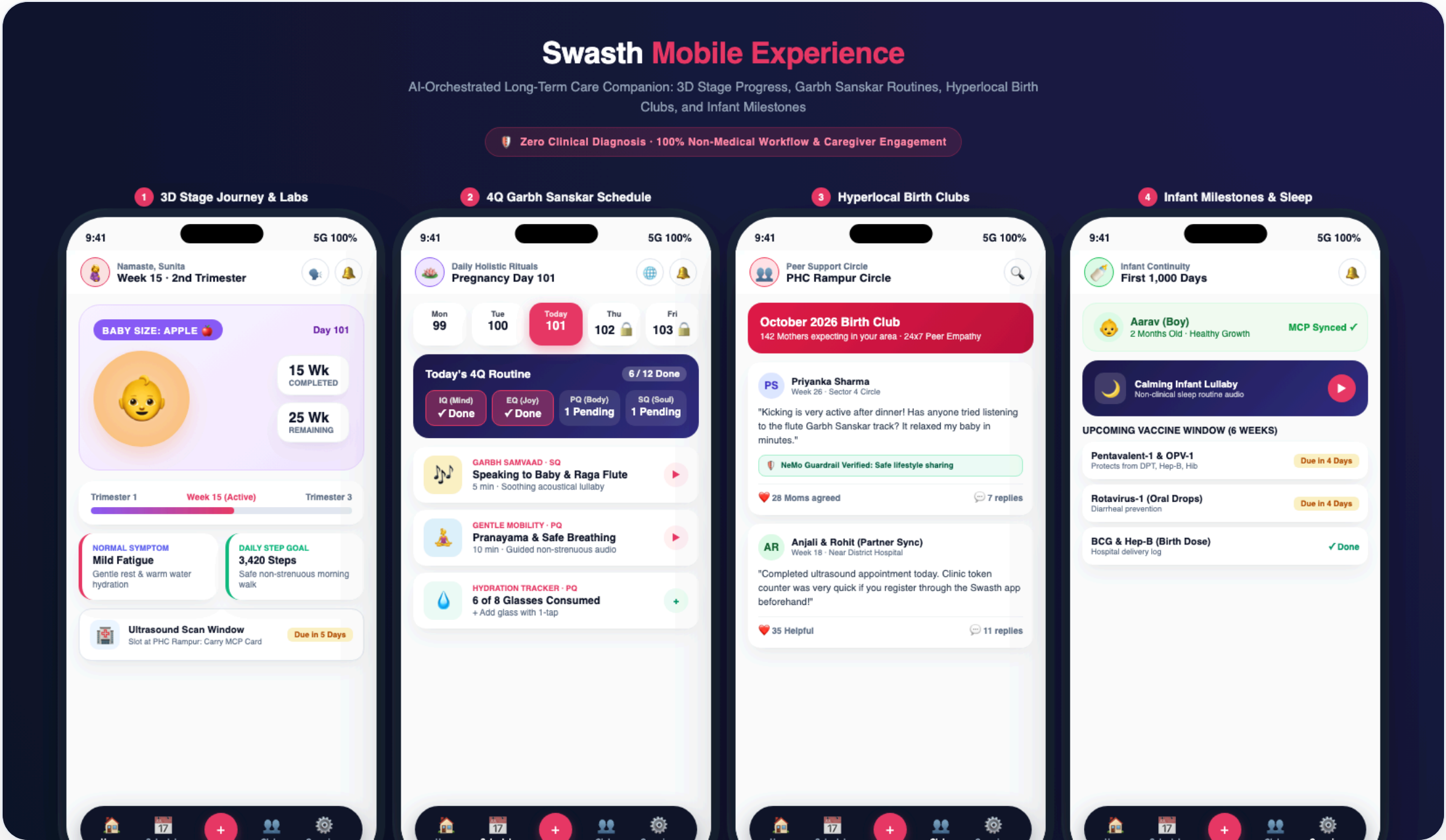
Target Metric

Zero Lost Health Records

Swasth · Strategic Hackathon Pillars

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App Design Showcase: Daily Rituals, Tools & Birth Clubs



1. Daily 4Q Schedule (Garbh Sanskar)

Structured tracking across IQ (cognitive), EQ (emotional), PQ (physical mobility), and SQ (spiritual) with 12 daily micro-activities.

2. Smart Tools & MCP Scanner

1-tap pastel tool grid: Pregnancy Tracker, Water Calculator, Steps, Om Chanter, MCP Card Photo Scanner, and Scheme Benefits.

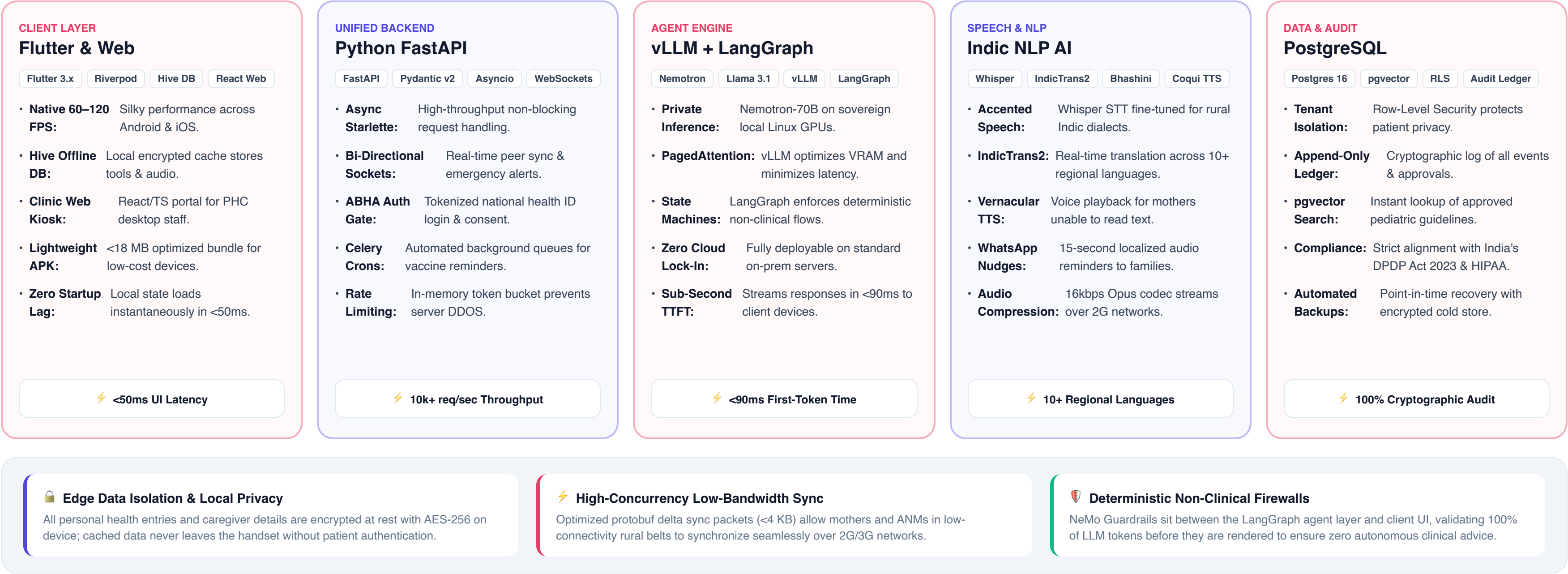
3. Hyperlocal Birth Clubs & Community

Peer circles connecting mothers in the same PHC radius, with AI guardrails protecting against unsafe medical home advice.

4. Vaccine Cohorts & Weaning Diet

Automated 6/10/14-week immunization schedules and doctor-approved month-by-month infant food meal plans.

Enterprise Open-Source Architecture: Speed & Reliability



Clinical Safety: Zero Diagnostic Risk & Guardrail Moderation

Strict Non-Clinical Scope

MANDATORY BOUNDARY

✗ 0% Diagnostic Advice:

Never infers medical diseases, pregnancy risk scores, or clinical conditions.

✗ 0% Prescriptions:

Strictly prohibits suggesting pharmaceutical drugs, doses, or medication edits.

✗ 0% Lab Interpretation:

Never attempts to interpret ultrasound Doppler scans or blood work numbers.

✓ Operational Assistive Only:

Strictly constrained to scheduling, diet charts, and welfare schemes.

📄 Persistent Disclaimers:

Clear advisory banner visible on every interaction and notification.

🛡️ Zero Clinical Liability · Strictly Non-Diagnostic

NVIDIA NeMo Guardrails

AUTOMATED AI DEFENSE

🛡️ In-Memory Semantic Firewall:

Dual-pass validation on both user prompts and model responses.

🚨 Red-Flag Interception:

Traps critical keywords (bleeding, acute pain, high fever, fluid loss).

🛑 Instant AI Halt:

Freezes non-clinical chat when an acute medical emergency is suspected.

☎️ 108 / PHC Fast-Routing:

Immediately displays 1-tap emergency dialer with PHC GPS coordinates.

🚫 Misinformation Filter:

Blocks harmful folk remedies and unverified claims in Birth Clubs.

🌐 Real-Time Semantic Interception Powered by Colang

Human-in-the-Loop & Audit

CLINICAL GOVERNANCE

👤 ANM & ASHA Oversight:

Frontline health workers review milestone due-lists with 1-click approvals.

📖 Immutable PostgreSQL Ledger:

Cryptographic audit trail of all alerts, overrides, and actions.

🔒 Granular Caregiver Consent:

Family syncing requires explicit maternal opt-in under DPDP 2023.

📊 Clinical Periodic Review:

Monthly physician committee audit of conversational logs & queries.

⚡ Emergency Fallback:

If offline, surfaces hardcoded directory of local PHC numbers.

👥 Frontline Worker Empowerment · 100% Auditable

🔄 Real-Time Guardrail Triage & Red-Flag Escalation Workflow

STEP 1 · INPUT

Vernacular Prompt

Mother inputs text/audio in regional language inquiring about routine nutrition or symptoms.

STEP 2 · NEMO GUARD

Semantic Inspection

In-memory Colang policy parses intent against clinical symptom taxonomy and dangerous keywords.

STEP 3A · NORMAL PATH

Assistive Guidance

If safe, responds with vetted diet tips, Garbh Sanskar routines, or vaccine schedule timelines.

STEP 3B · RED FLAG

Emergency Routing

If red flag detected, automation immediately ceases; surfaces 108 dialer and on-duty PHC nurse.

Swasth · Safety & Governance

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Measurable Operational Outcomes within 60–90 Days

85% Drop

Frontline Admin Burden

Reduces manual paper due-list compilation from 10 hrs to <1.5 hrs/week per ANM.

90%+

Vaccine Adherence

Cuts 6 to 14-week infant immunization dropouts via localized voice reminders.

4x Higher

Family & Peer Activity

Birth Clubs & caregiver WhatsApp synchronization drive community support.

100%

Auditable Compliance

Zero unreviewed clinical actions with complete append-only event logging.

Execution & Rollout Roadmap (60–90 Days)

DAY 1 – 30

MVP & Guardrail Validation

Deploy Flutter mobile client + Python backend; benchmark open-source Nemotron on vLLM; validate NeMo non-clinical safety boundaries.

DAY 31 – 60

Pilot with 5 PHCs & Birth Clubs

Onboard 500 expectant mothers & caregivers; launch 5 localized Birth Clubs; track daily Garbh Sanskar engagement and vaccine adherence.

DAY 61 – 90

Scale & HMIS Integration

Synchronize with state RCH registers; export verified administrative audit reports; expand vernacular voice models into 10+ dialects.